TRIGONOMETRY & COMPLEX NUMBERS

PART 1: TRIGONOMETRY

1. Trigonometric Functions & Angles of Any Magnitude

- **Deep Definition:** Trigonometric functions ($\sin, \cos, \tan, \csc, \sec, \cot$) relate the angles of a right-angled triangle to ratios of its side lengths. For angles of *any magnitude* (positive, negative, or greater than 360°), we use the **Cartesian Coordinate System (Unit Circle)**.
- **The ASTC Rule (All Silver Tea Cups / Quadrant Signs):**
  - **1st Quadrant (0° to 90° / 0 to $\pi/2$): All** trigonometric ratios are positive.
  - **2nd Quadrant (90° to 180° / $\pi/2$ to π):** Only **Sine** ($\sin$) and Cosec ($\csc$) are positive.
  - **3rd Quadrant (180° to 270° / π to $3\pi/2$):** Only **Tangent** ($\tan$) and Cotangent ($\cot$) are positive.
  - **4th Quadrant (270° to 360° / $3\pi/2$ to 2π):** Only **Cosine** ($\cos$) and Secant ($\sec$) are positive.
- **Solved Question:** Find the exact value of $\sin(765^\circ)$.
  - **Solution:**
    1. Divide 765° by 360° to find the number of full rotations:
       $765^\circ = 2 \times 360^\circ + 45^\circ$
    2. This means $\sin(765^\circ)$ is equivalent to $\sin(45^\circ)$ in the 1st quadrant.
    3. Therefore, $\sin(765^\circ) = \sin(45^\circ) = \frac{1}{\sqrt{2}}$. **(Answer)**

2. Compound and Multiple Angles

- **Compound Angle Formulas (Addition/Subtraction):**
  - $\sin(A + B) = \sin A \cos B + \cos A \sin B$
  - $\sin(A - B) = \sin A \cos B - \cos A \sin B$
  - $\cos(A + B) = \cos A \cos B - \sin A \sin B$
  - $\cos(A - B) = \cos A \cos B + \sin A \sin B$
  - $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$
- **Multiple Angle Formulas (Double Angle Formulas):**
  - $\sin 2A = 2 \sin A \cos A = \frac{2\tan A}{1+\tan^2 A}$
  - $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A = \frac{1-\tan^2 A}{1+\tan^2 A}$
  - $\tan 2A = \frac{2\tan A}{1-\tan^2 A}$
- **Solved Question:** Prove that $\frac{\sin 2A}{1+\cos 2A} = \tan A$.
  - **Solution:**
    - Substitute double angle formulas into the LHS:
      - $\sin 2A = 2 \sin A \cos A$
      - $\cos 2A = 2 \cos^2 A - 1 \implies 1 + \cos 2A = 2 \cos^2 A$
    - LHS = $\frac{2\sin A \cos A}{2\cos^2 A}$
    - Cancel common terms (2 and $\cos A$):
      LHS = $\frac{\sin A}{\cos A} = \tan A = RHS$. **(Hence Proved)**

3. Inverse Circular Functions

- **Deep Definition:** Inverse trigonometric functions ($\sin^{-1}x, \cos^{-1}x, \tan^{-1}x$, etc.) are the inverse operations of the standard trigonometric ratios. They return the angle whose trigonometric ratio is given.

- **Important Properties & Ranges:**
  - $\sin^{-1}(\sin x) = x$ for $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$
  - $\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$
  - $\sin^{-1}(-x) = -\sin^{-1}x$
  - $\cos^{-1}(-x) = \pi - \cos^{-1}x$
- **Solved Question:** Find the principal value of $\sin^{-1}\left(-\frac{1}{2}\right) + \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$.
  - **Solution:**
    1. Let $\theta_1 = \sin^{-1}\left(-\frac{1}{2}\right)$. We know $\sin(\pi/6) = 1/2$, so since $\sin^{-1}(-x) = -\sin^{-1}x$: $\theta_1 = -\frac{\pi}{6}$ (which lies in the principal range $[-\pi/2, \pi/2]$).
    2. Let $\theta_2 = \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$. Since $\cos(\pi/6) = \sqrt{3}/2$ and $\cos^{-1}(-x) = \pi - \cos^{-1}x$: $\theta_2 = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$.
    3. Sum = $-\frac{\pi}{6} + \frac{5\pi}{6} = \frac{4\pi}{6} = \frac{2\pi}{3}$. **(Answer)**

PART 2: COMPLEX NUMBERS

1. Introduction, Algebraic Operations, & Complex Conjugate

- **Deep Definition:** A complex number is a number of the form $z = x + iy$, where x is the **Real part** ($Re(z)$), y is the **Imaginary part** ($Im(z)$), and i is the imaginary unit satisfying $i^2 = -1$.
- **Complex Conjugate:** If $z = x + iy$, its complex conjugate is denoted by $\bar{z} = x - iy$. (Property: $z \cdot \bar{z} = x^2 + y^2$).
- **Algebraic Operations Rules:**
  - **Addition:** $(a + ib) + (c + id) = (a + c) + i(b + d)$
  - **Multiplication:** $(a + ib)(c + id) = (ac - bd) + i(ad + bc)$
  - **Division:** Multiply numerator and denominator by the conjugate of the denominator.
- **Solved Question:** Simplify $\frac{2+3i}{1-2i}$ and express it in standard $x + iy$ form.
  - **Solution:**
    1. Multiply numerator and denominator by the conjugate of $(1 - 2i)$, which is $(1 + 2i)$:

       2. $\frac{(2+3i)(1+2i)}{(1-2i)(1+2i)}$
    3. Expand numerator:
       $2(1) + 2(2i) + 3i(1) + 3i(2i) = 2 + 4i + 3i + 6i^2$
       Since $i^2 = -1$, this becomes: $2 + 7i - 6 = -4 + 7i$.
    4. Expand denominator using $(a - b)(a + b) = a^2 - b^2$:
       $1^2 - (2i)^2 = 1 - 4i^2 = 1 - 4(-1) = 1 + 4 = 5$.
    5. Combine:

       6. $\frac{-4+7i}{5} = -\frac{4}{5} + i\frac{7}{5}$ ($x = -4/5, y = 7/5$)
    7. **(Answer)**

2. Modulus, Argument, Polar Form, & Vector Form

- **Modulus (r):** The distance of the complex point from the origin: $\vert{}z\vert{} = r = \sqrt{x^2 + y^2}$.
- **Argument (θ):** The angle made with the positive real axis: $\theta = \tan^{-1}\left(\frac{y}{x}\right)$.
- **Polar Form:** $z = r(\cos\theta + i\sin\theta) = re^{i\theta}$ (Euler's Form).

- **Vector Form:** A complex number $z = x + iy$ can be represented geometrically as a position vector (x, y) in the 2D Argand plane.
- **Solved Question:** Find the modulus, argument, and polar form of the complex number $z = -1 + i\sqrt{3}$.
    - **Solution:**
        1. **Modulus (r):**

        2. $$r = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{1 + 3} = \sqrt{4} = 2$$
        3. **Argument (θ):**

           Here $x = -1$ (negative) and $y = \sqrt{3}$ (positive). This lies in the **2nd Quadrant**.
           $$\alpha = \tan^{-1}\left\vert{}\frac{\sqrt{3}}{-1}\right\vert{} = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$$

        4. $$\theta = \pi - \alpha = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$
        5. **Polar Form:**

        6. $$z = 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$$
        7. **(Answer)**

3. De-Moivre's Theorem & Roots of a Complex Number

- **De-Moivre's Theorem Statement:** For any real number n,
    - $(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$
- **Finding Roots of a Complex Number:** To find the n-th roots of a complex number

$$z^{1/n} = r^{1/n}\left[\cos\left(\frac{\theta + 2k\pi}{n}\right) + i\sin\left(\frac{\theta + 2k\pi}{n}\right)\right] \quad \text{for } k = 0, 1, 2, \ldots, n-1$$

- **Solved Question:** Evaluate $(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})^8$ using De-Moivre's theorem.
    - **Solution:**
        1. Apply De-Moivre's Theorem by multiplying the angle by the exponent $n = 8$:

        2. $$\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)^8 = \cos\left(8 \times \frac{\pi}{4}\right) + i\sin\left(8 \times \frac{\pi}{4}\right)$$
        3. Simplify the angles:

        4. $$= \cos(2\pi) + i\sin(2\pi)$$
        5. Evaluate standard trigonometric values:
           $\cos(2\pi) = 1$ and $\sin(2\pi) = 0$.
        6. Final Result: $1 + i(0) = 1$. **(Answer)**

MATRICES AND DETERMINANTS

1. Introduction & Types of Matrices

- **Deep Definition:** A matrix is a rectangular array or table of numbers arranged in rows (horizontal lines) and columns (vertical lines). The dimensions of a matrix are given as $m \times n$ (m rows and n columns).

Types of Matrices:

1. **Row Matrix:** A matrix having only one row ($1 \times n$). Example: $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$.
2. **Column Matrix:** A matrix having only one column ($m \times 1$). Example: $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$.
3. **Square Matrix:** A matrix where the number of rows equals the number of columns ($m = n$).
4. **Diagonal Matrix:** A square matrix where all non-diagonal elements are zero ($a_{ij} = 0$ for $i \neq j$).
5. **Scalar Matrix:** A diagonal matrix in which all diagonal elements are equal.
6. **Identity (Unit) Matrix (I):** A diagonal matrix where all diagonal elements are equal to 1.
7. **Null (Zero) Matrix:** A matrix where every element is zero (0).
8. **Upper/Lower Triangular Matrix:** A square matrix where all elements below/above the main diagonal are zero.
9.

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2. Algebraic Operations, Transpose, & Special Matrices

A. Algebraic Operations

- **Addition/Subtraction:** Can only be performed on matrices of the *same order*. Performed element-by-element: $(A \pm B)_{ij} = a_{ij} \pm b_{ij}$.
- **Scalar Multiplication:** Multiplying every element of a matrix by a scalar constant k: $(kA)_{ij} = k \cdot a_{ij}$.
- **Matrix Multiplication:** If A is of order $m \times n$ and B is of order $n \times p$, their product $C = AB$ (order $m \times p$) is obtained by taking the dot product of rows of A and columns of B:
    - $$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$
- *(Note: Matrix multiplication is **not commutative** i.e., $AB \neq BA$ in general).*

B. Transpose of a Matrix (A^T or A')

- **Definition:** Interchanging the rows and columns of a matrix. If $A = [a_{ij}]_{m \times n}$, then $A^T = [a_{ji}]_{n \times m}$.
- **Properties of Transpose:**

1. $(A^T)^T = A$
2. $(A + B)^T = A^T + B^T$
3. $(AB)^T = B^T A^T$ (Reversal Law)

C. Symmetric and Skew-Symmetric Matrices

- **Symmetric Matrix:** A square matrix A is symmetric if $A^T = A$ (implies $a_{ij} = a_{ji}$).
- **Skew-Symmetric Matrix:** A square matrix A is skew-symmetric if $A^T = -A$ (implies $a_{ij} = -a_{ji}$, and all diagonal elements must be zero).
- *Key Theorem:* Any square matrix can be expressed uniquely as the sum of a symmetric and a skew-symmetric matrix:

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$$

3. Orthogonal Matrices & Determinants

A. Orthogonal Matrix

- **Definition:** A square matrix A of order n is called an **orthogonal matrix** if its transpose is equal to its inverse:

$$A \cdot A^T = A^T \cdot A = I (or A^{-1} = A^T)$$

- **Determinant Property:** The determinant of an orthogonal matrix is always ± 1 (i.e., $\vert{}A\vert{} = \pm 1$).

B. Determinant of a Square Matrix

- **Definition:** A scalar value computed exclusively from a square matrix.
-
  - For a 2×2 Matrix ($A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$):
    $$|A| = ad - bc$$
  - **For a 3×3 Matrix:** Evaluated by expanding along any row or column using cofactor minors.

4. Inverse of a Square Matrix & Rank of a Matrix

A. Inverse of a Square Matrix (A^{-1})

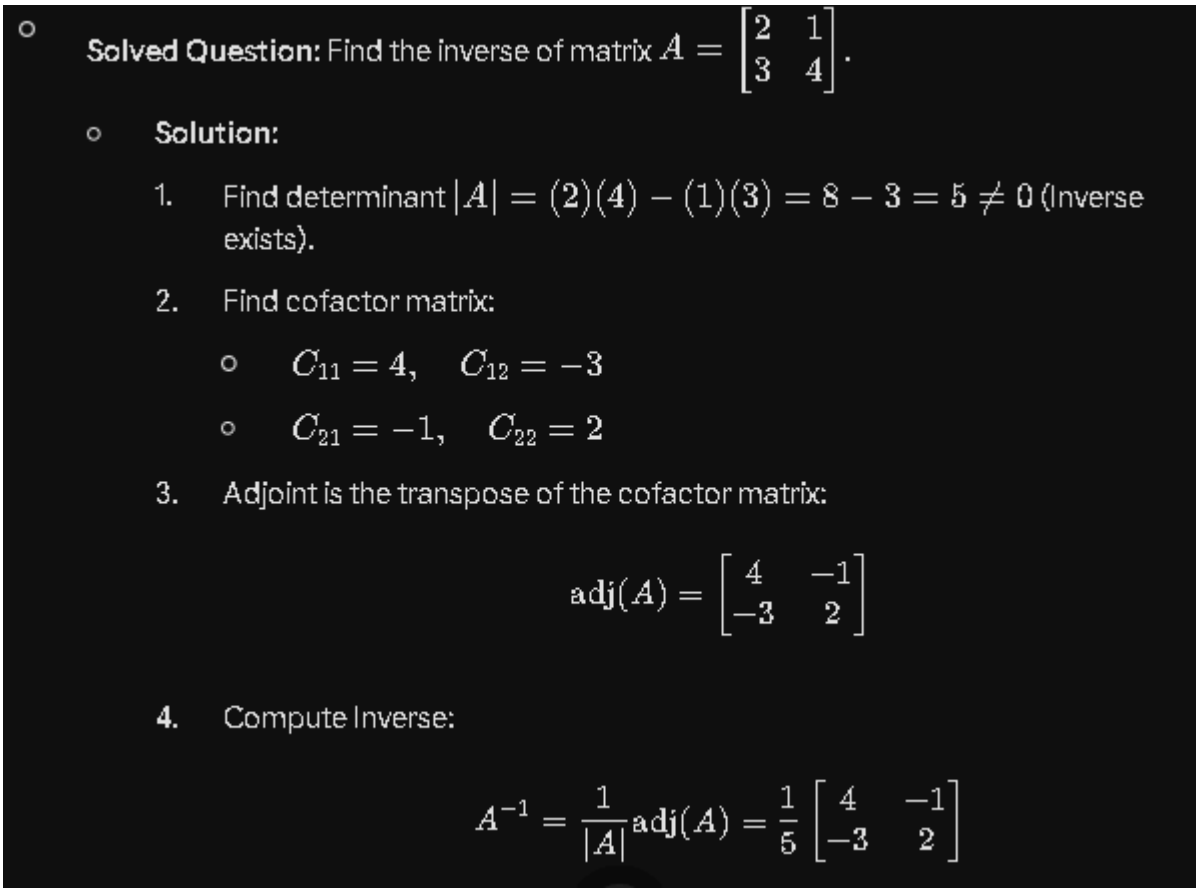
- **Definition:** If A is a non-singular square matrix ($\vert{}A\vert{} \neq 0$), its inverse A^{-1} exists such that:

$$A \cdot A^{-1} = A^{-1} \cdot A = I$$

- **Formula using Adjoint:**

$$A^{-1} = \frac{\text{adj}(A)}{|A|}$$

- **Solved Question:**
- Find the inverse of matrix $A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$.
- **Solution:**

1. 

B. Rank of a Matrix

- **Definition:** The rank of a matrix is the maximum number of linearly independent row (or column) vectors in the matrix. It is determined by reducing the matrix to its **Row Echelon Form** using elementary row operations. The rank equals the number of non-zero rows remaining.

5. Solution of Linear Equations

A. Cramer's Rule (Determinant Method)

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For a system of two linear equations:

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

- **Formulas:**

$$x = \frac{D_x}{D}, \quad y = \frac{D_y}{D}$$

where:

$$D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}, \quad D_x = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}, \quad D_y = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$$

1.

- **Solved Question:** Solve using Cramer's Rule: $2x + 3y = 7$ and $3x - 2y = 4$.
  - **Solution:**
    1. Compute determinant D:

$$D = \begin{vmatrix} 2 & 3 \\ 3 & -2 \end{vmatrix} = (2)(-2) - (3)(3) = -4 - 9 = -13$$

    2. Compute D_x (replace 1st column with constants $7, 4$):

$$|D_x = \begin{vmatrix} 7 & 3 \\ 4 & -2 \end{vmatrix} = (7)(-2) - (3)(4) = -14 - 12 = -26$$

    3. Compute D_y (replace 2nd column with constants $7, 4$):

$$D_y = \begin{vmatrix} 2 & 7 \\ 3 & 4 \end{vmatrix} = (2)(4) - (7)(3) = 8 - 21 = -13$$

    4. Find values of x and y:

$$x = \frac{D_x}{D} = \frac{-26}{-13} = 2, \quad y = \frac{D_y}{D} = \frac{-13}{-13} = 1$$

    Result: $x = 2, y = 1$. **(Answer)**

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B. Gauss-Elimination Method

- **Definition:** A systematic method using elementary row transformations to convert an augmented matrix $[A|B]$ into an **Upper Triangular Matrix**, followed by back-substitution to find the variables.

6. Eigenvalues and Eigenvectors of a Square Matrix

- **Characteristic Equation:** For a square matrix A, the eigenvalues λ are roots of the characteristic polynomial:
  - $\det(A - \lambda I) = 0$
- **Eigenvectors:** For each eigenvalue λ, the corresponding eigenvector X is found by solving the homogeneous matrix equation:
  - $(A - \lambda I)X = 0$
- **Solved Question:** Find the eigenvalues of the matrix $2x + 3y = 7$ and $3x - 2y = 4$.
- **Solution:**

- Solved Question: Find the eigenvalues of the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$.
  - Solution:
    1. Set up the characteristic equation $\det(A - \lambda I) = 0$:

$$\det\left(\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = 0$$

$$\det\begin{bmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{bmatrix} = 0$$

    2. Expand the determinant:

$$(1-\lambda)(2-\lambda) - (2)(3) = 0$$

$$(2 - \lambda - 2\lambda + \lambda^2) - 6 = 0$$

$$\lambda^2 - 3\lambda - 4 = 0$$

    3. Factorize the quadratic equation:

$$(\lambda - 4)(\lambda + 1) = 0$$

    4. Solve for eigenvalues:

$$\lambda_1 = 4, \quad \lambda_2 = -1$$

    (Answer)

MASTERCLASS: EIGENVALUES & EIGENVECTORS OF A 3x3 MATRIX

1. Step-by-Step Algorithm

1. **Characteristic Equation:** Matrix A ke eigenvalues (λ) nikalne ke liye formula use karte hain:

$$\det(A - \lambda I) = 0$$

Isse ek cubic polynomial equation banti hai ($\lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0$), jisme S_1 trace hai aur S_2 sum of principal minors hai.

2. **Find Eigenvalues (λ):** Cubic equation ko solve karke 3 roots ($\lambda_1, \lambda_2, \lambda_3$) nikalte hain.

3. **Find Eigenvectors (X):** Har ek eigenvalue λ ke liye, matrix equation $(A - \lambda I)X = 0$ ko solve karke corresponding eigenvector $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ nikalte hain.

2. Fully Solved Practice Numerical Problem

Problem: Find the eigenvalues and the corresponding eigenvectors for the 3×3 matrix A:

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Step 1: Find the Characteristic Equation

Formulate $\det(A - \lambda I) = 0$:

$$\det \begin{bmatrix} 2-\lambda & 1 & 1 \\ 1 & 2-\lambda & 1 \\ 1 & 1 & 2-\lambda \end{bmatrix} = 0$$

Is determinant ko expand karne ke liye shortcut properties ya standard expansion use karte

hain. Is matrix ke liye characteristic polynomial hoti hai:

$$\lambda^3 - (\text{Trace})\lambda^2 + (\text{Sum of Principal Minors})\lambda - \det(A) = 0$$

- $\text{Trace} = 2 + 2 + 2 = 6$
- Principal Minors = $\begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = (3) + (3) + (3) = 9$
- $\det(A) = 2(4-1) - 1(2-1) + 1(1-2) = 2(3) - 1(1) + 1(-1) = 6 - 1 - 1 = 4$

To equation bani:

$$\lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0$$

Step 2: Solve for Eigenvalues (λ)

Trial and error method se dekhte hain: agar $\lambda = 1$ rakhein:

$$(1)^3 - 6(1)^2 + 9(1) - 4 = 1 - 6 + 9 - 4 = 0$$

Matlab $\lambda = 1$ ek root hai. Ab synthetic division ya factor theorem se polynomial factorize karenge:

$$(\lambda - 1)(\lambda^2 - 5\lambda + 4) = 0$$

Quadratic part ko factorize karo: $(\lambda - 1)(\lambda - 1)(\lambda - 4) = 0$.

- **Eigenvalues:** $\lambda = 1, 1, 4$ (Yahan $\lambda = 1$ repeated root hai).

Step 3: Find Eigenvectors for each λ

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Case A: For $\lambda = 4$ (Distinct Eigenvalue)

Matrix $(A - 4I)X = 0$ likho:

$$\begin{bmatrix} 2-4 & 1 & 1 \\ 1 & 2-4 & 1 \\ 1 & 1 & 2-4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Equations bani:

1. $-2x_1 + x_2 + x_3 = 0$
2. $x_1 - 2x_2 + x_3 = 0$

Cross-multiplication method (or ratio method) use karo coefficients par:

$$\frac{x_1}{1-(-2)} = \frac{x_2}{1-(-2)} = \frac{x_3}{4-1} \implies \frac{x_1}{3} = \frac{x_2}{3} = \frac{x_3}{3}$$

Simplify karke ratio mila: $x_1 : x_2 : x_3 = 1 : 1 : 1$.

Simplify karke ratio mila: $x_1 : x_2 : x_3 = 1 : 1 : 1$.

- First Eigenvector (X_1): $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

Case B: For $\lambda = 1$ (Repeated Eigenvalue)

Matrix $(A - 1I)X = 0$ likho:

$$\begin{bmatrix} 2-1 & 1 & 1 \\ 1 & 2-1 & 1 \\ 1 & 1 & 2-1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Teeno rows ek hi equation deti hain:

$$x_1 + x_2 + x_3 = 0$$

Yahan sirf ek hi independent equation mili jabki variables 3 hain. Iska matlab hume do free variables assume karne padenge ($x_2 = s, x_3 = t$):

- Let $x_2 = s$ and $x_3 = t$, toh $x_1 = -s - t$.
- Vector form:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -s-t \\ s \\ t \end{bmatrix} = s\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Isse hame do linearly independent eigenvectors mil gaye:

- **Second Eigenvector (X_2):** $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$
- **Third Eigenvector (X_3):** $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

Final Answer Summary:

- **Eigenvalues:** $\lambda = 4, 1, 1$
- **Corresponding Eigenvectors:**

$$X_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad X_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad X_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

DIFFERENTIAL CALCULUS

PART 1: LIMIT, CONTINUITY, AND DIFFERENTIABILITY

1. Limits

- **Deep Definition:** The limit of a function $f(x)$ as x approaches a value c is the value that $f(x)$ gets closer to as x gets closer and closer to c (from both left and right sides).
  - $\lim_{x \to c} f(x) = L$
- **Standard Limit Formulas:**
  - $\lim_{x \to 0} \frac{\sin x}{x} = 1$

- $\lim_{x \to 0} \frac{e^x - 1}{x} = 1$
- $\lim_{x \to 0} \frac{\ln(1+x)}{x} = 1$
- $\lim_{x \to a} \frac{x^n - a^n}{x - a} = n \cdot a^{n-1}$

- **Solved Question:** Evaluate $\lim_{x \to 0} \frac{\sin 3x}{2x}$.
  - **Solution:**
    - $\lim_{x \to 0} \frac{\sin 3x}{2x} = \frac{1}{2} \cdot \lim_{x \to 0} \left(\frac{\sin 3x}{3x} \cdot 3 \right) = \frac{3}{2} \cdot (1) = \frac{3}{2}$
  - **(Answer)**

2. Continuity

- **Deep Definition:** A function $f(x)$ is continuous at a specific point $x = c$ if there is no break, jump, or hole in the graph at that point. Mathematically, three conditions must hold simultaneously:
  1. $f(c)$ must be defined.
  2. $\lim_{x \to c} f(x)$ must exist (i.e., $LHL = RHL$).
  3. $\lim_{x \to c} f(x) = f(c)$.

3. Differentiability

- **Deep Definition:** A function is differentiable at $x = c$ if its derivative exists at that point, which means the left-hand derivative equals the right-hand derivative:
  - $f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h}$
- *(Note: Every differentiable function is continuous, but a continuous function is not necessarily differentiable—e.g., sharp corners like $f(x) = \vert x \vert$ at $x = 0$).*

PART 2: DIFFERENTIATION RULES & STANDARD FUNCTIONS

1. Core Differentiation Rules

- **Sum/Difference Rule:** $\frac{d}{dx}[u(x) \pm v(x)] = \frac{du}{dx} \pm \frac{dv}{dx}$
- **Product Rule:** $\frac{d}{dx}[u \cdot v] = u\frac{dv}{dx} + v\frac{du}{dx}$
- **Quotient Rule:** $\frac{d}{dx}\left[\frac{u}{v}\right] = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$
- **Chain Rule (Composite Functions):** $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

2. Derivatives of Standard Functions

- **Algebraic:** $\frac{d}{dx}(x^n) = nx^{n-1}$
- **Exponential:** $\frac{d}{dx}(e^x) = e^x$, $\frac{d}{dx}(a^x) = a^x \ln a$
- **Logarithmic:** $\frac{d}{dx}(\ln x) = \frac{1}{x}$
- **Trigonometric:**
  - $\frac{d}{dx}(\sin x) = \cos x$
  - $\frac{d}{dx}(\cos x) = -\sin x$

- $\frac{d}{dx}(\tan x) = \sec^2 x$
- **Inverse Trigonometric:**
  - $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$
  - $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$
- **Solved Question:** Find $\frac{dy}{dx}$ if $y = x^3 \cdot \ln x$.
  - **Solution:** Apply the Product Rule ($u = x^3$, $v = \ln x$):
    - $\frac{dy}{dx} = x^3 \cdot \frac{d}{dx}(\ln x) + \ln x \cdot \frac{d}{dx}(x^3)$
    - $\frac{dy}{dx} = x^3\left(\frac{1}{x}\right) + \ln x \cdot (3x^2) = x^2 + 3x^2 \ln x = x^2(1 + 3\ln x)$
  - **(Answer)**

PART 3: TANGENT AND NORMAL LINES & CONDITION OF TANGENCY

1. Equations of Tangent and Normal

- **Derivative as Slope:** The derivative $\frac{dy}{dx}$ evaluated at a point (x_1, y_1) represents the slope (m) of the tangent line to the curve at that point.
- **Equation of Tangent:**
  - $y - y_1 = m(x - x_1)$
- **Equation of Normal:** (Since the normal is perpendicular to the tangent, its slope is $-\frac{1}{m}$):
  - $y - y_1 = -\frac{1}{m}(x - x_1)$
- **Solved Question:** Find the equation of the tangent line to the curve $y = x^2 - 3x + 2$ at the point where $x = 2$.
  - **Solution:**
    1. Find the y-coordinate when $x = 2$:
    2. $y_1 = (2)^2 - 3(2) + 2 = 4 - 6 + 2 = 0$
    3. Point of contact is $(2, 0)$.
    4. Find the derivative $\frac{dy}{dx}$ to get the slope m:
    5. $\frac{dy}{dx} = 2x - 3$
    6. Evaluate slope at $x = 2$:
    7. $m = 2(2) - 3 = 4 - 3 = 1$
    8. Apply the tangent equation $y - y_1 = m(x - x_1)$:

       $$y - 0 = 1(x - 2) \implies y = x - 2$$

       (Answer)

PART 4: EXTREME VALUES OF FUNCTIONS (MAXIMA AND MINIMA)

1. Second Derivative Test for Extrema

To find the maximum or minimum values of a function $f(x)$:

1. **Step 1:** Find the first derivative $f'(x)$ and set it to zero ($f'(x) = 0$) to find **critical points**.
2. **Step 2:** Find the second derivative $f''(x)$.
3. **Step 3:** Evaluate $f''(x)$ at each critical point:
    - If $f''(c) < 0$, the function has a **Local Maximum** at $x = c$.
    - If $f''(c) > 0$, the function has a **Local Minimum** at $x = c$.
    - If $f''(c) = 0$, the test fails (further analysis needed).

- **Solved Question:** Find the local extrema of the function $f(x) = 2x^3 - 9x^2 + 12x + 5$.
    - **Solution:**
        1. Find first derivative $f'(x)$:

        2. $f'(x) = 6x^2 - 18x + 12$
        3. Set $f'(x) = 0$ for critical points:
           $$6(x^2 - 3x + 2) = 0 \implies (x - 1)(x - 2) = 0$$
           Critical points are $x = 1$ and $x = 2$.
        4. Find second derivative $f''(x)$:

        5. $f''(x) = 12x - 18$
        6. Test at $x = 1$:

        7. $f''(1) = 12(1) - 18 = -6 < 0 (Local Maximum at x = 1)$
        8. Test at $x = 2$:

        9. $f''(2) = 12(2) - 18 = 6 > 0 (Local Minimum at x = 2)$

(Answer)

ADVANCED INTEGRAL CALCULUS & MULTIPLE INTEGRALS

1. Indefinite Integrals & Basic Formulae

- **Deep Definition: An indefinite integral represents a family of functions whose derivative is the given integrand. If $\frac{d}{dx}[F(x)] = f(x)$, then:**

    - $\int f(x)dx = F(x) + C$

- **where C is the arbitrary constant of integration, accounting for any vertical shifts in the original function.**

Core Standard Formulae:

- $\int x^n dx = \frac{x^{n+1}}{n+1} + C (n \neq -1)$
- **$\int \frac{1}{x} \, dx = \ln\vert{}x\vert{} + C$**
- $\int e^x dx = e^x + C$

- $\int \sin x dx = -\cos x + C$
- $\int \sec^2 x dx = \tan x + C$

2. Integration by Substitution

- **Deep Definition: Also known as the u-substitution method, this technique is the reverse of the chain rule in differentiation. It is used when an integrand contains a composite function alongside its derivative factor.**
- **Algorithm:**
  - **Choose a substitution variable $u = g(x)$.**
  - **Compute the differential $du = g'(x)dx$.**
  - **Substitute u and du into the integral to transform it into a standard, easily solvable form.**
  - **Integrate with respect to u and substitute back the original variable x.**
- **Solved Numerical: Evaluate $\int 2x \cos(x^2)dx$.**
  - **Solution:**
    1. **Let $u = x^2$.**
    2. **Differentiate: $du = 2xdx$.**
    3. **Substitute into the integral:**
    4. $\int \cos(u)du$
    5. **Integrate $\cos(u)$ with respect to u:**
    6. $= \sin(u) + C$
    7. **Substitute back $u = x^2$:**
    8. $= \sin(x^2) + C$
    9. **(Answer)**

3. Integration by Parts

- **Deep Definition: Derived from the product rule of differentiation, integration by parts is used to integrate the product of two distinct functions.**
- **Formula:**
  - $\int u \cdot v dx = u\int v dx - \int\left(\frac{du}{dx}\int v dx\right)dx$
- **The ILATE Rule for choosing u:**
  - **Inverse trigonometric, Logarithmic, Algebraic, Trigonometric, Exponential.**
- **Solved Numerical: Evaluate $\int xe^x dx$.**
  - **Solution:**
    1. **Choose $u = x$ (Algebraic) and $v = e^x$ (Exponential).**
    2. **Compute $\frac{du}{dx} = 1$ and $\int v dx = e^x$.**
    3. **Apply the formula:**

4. $\int xe^x dx = xe^x - \int (1 \cdot e^x) dx$

5.

6. $= xe^x - e^x + C = e^x(x - 1) + C$

7. **(Answer)**

4. Definite Integrals & Properties of Definite Integrals

- **Deep Definition: A definite integral has upper and lower limits (a and b), representing the net accumulated signed area under a curve.**

$$\bullet \quad \int_a^b f(x)dx = F(b) - F(a)$$

Important Properties of Definite Integrals:

1. **Reflection Property:** $\int_a^b f(x)dx = \int_a^b f(a + b - x)dx$
2. **King's Property:** $\int_a^b f(x)dx = \int_a^b f(a + b - x)dx$
3. **Symmetry Property:**
   - **If $f(x)$ is an even function ($f(-x) = f(x)$):** $\int_{-a}^{a} f(x)dx = 2\int_0^a f(x)dx$
   - **If $f(x)$ is an odd function ($f(-x) = -f(x)$):** $\int_{-a}^{a} f(x)dx = 0$

- **Solved Numerical: Evaluate $\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx$ using properties.**
  - **Solution:**
    1. **Let $I = \int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx$ --- (Equation 1)**
    2. **Apply the reflection property $\int_a^b f(x)dx = \int_a^b f(a + b - x)dx$:**

    3. $I = \int_0^{\pi/2} \frac{\sin\left(\frac{\pi}{2} - x\right)}{\sin\left(\frac{\pi}{2} - x\right) + \cos\left(\frac{\pi}{2} - x\right)} dx$
    4.
    5. $I = \int_0^{\pi/2} \frac{\cos x}{\cos x + \sin x} dx$
    6. **--- (Equation 2)**
    7. **Add Equation 1 and Equation 2:**

    8. $2I = \int_0^{\pi/2} \frac{\sin x + \cos x}{\sin x + \cos x} dx = \int_0^{\pi/2} 1 dx$
    9.
    10. $2I = [x]_0^{\pi/2} = \frac{\pi}{2} - 0 = \frac{\pi}{2}$
    11. **Solve for I:**

    12. $I = \frac{\pi}{4}$

13. (Answer)

5. Evaluation of Double and Triple Integrals

A. Double Integrals

- **Deep Definition: Extends single integration to two dimensions over a region R in the xy-plane, computing volumes under surfaces or 2D areas.**
- **Evaluation via Iterated Integrals:**
  $$\iint_{R} f(x,y) \, dA = \int_{a}^{b} \int_{g_1(x)}^{g_2(x)} f(x,y) \, dy \, dx$$
- **Solved Numerical: Evaluate $\int_0^1 \int_0^2 (x + y) dy dx$.**
  - **Solution:**
    1. **Integrate with respect to y first (treating x as a constant):**

    2. $$\int_0^2 (x + y) dy = \left[xy + \frac{y^2}{2} \right]_{y=0}^{y=2} = (2x + 2) - (0) = 2x + 2$$
    3. **Integrate the resulting expression with respect to x:**

    4. $$\int_0^1 (2x + 2) dx = \left[x^2 + 2x \right]_0^1 = (1^2 + 2(1)) - (0) = 3$$
    5. **(Answer)**

B. Triple Integrals

- **Deep Definition: Extends integration to three-dimensional space over a volume V, used for calculating spatial mass, volume, and moments of inertia.**
  $$\iiint_{V} f(x,y,z) \, dV = \int_{x_1}^{x_2} \int_{y_1}^{y_2} \int_{z_1}^{z_2} f(x,y,z) \, dz \, dy \, dx$$

6. Applications of Definite Integrals: Area and Volume

A. Finding Area Under a Curve

The area A bounded by the curve $y = f(x)$, the x-axis, and vertical lines $x = a$ and $x = b$ is given by:

$$A = \int_a^b f(x) dx$$

- **Solved Numerical: Find the area bounded by the line $y = x$, the x-axis, and the ordinates $x = 0$ and $x = 4$.**
  - **Solution:**
  - $$A = \int_0^4 x dx = \left[\frac{x^2}{2} \right]_0^4 = \frac{4^2}{2} - 0 = \frac{16}{2} = 8 square units$$
  - **(Answer)**

B. Finding Volume of Revolution (Disk Method)

When a region bounded by $y = f(x)$ is revolved around the x-axis from $x = a$ to $x = b$, the volume V of the generated solid of revolution is:

$$V = \pi \int_a^b [f(x)]^2 dx$$

1. Fundamentals of Integration

- **Deep Definition: Integration is the reverse process of differentiation (hence also known as an antiderivative or primitive). If the derivative of a function $F(x)$ is $f(x)$, i.e., $F'(x) = f(x)$, then the integral of $f(x)$ with respect to x is $F(x) + C$, where C is the constant of integration.**
- **Indefinite Integral: An integral without upper and lower limits, representing a family of curves:**

  - $\int f(x)dx = F(x) + C$

- **Definite Integral: An integral evaluated between two specific limits (a and b), representing the net geometric area under a curve:**

  - $\int_a^b f(x)dx = F(b) - F(a)$

2. Standard Integration Formulas

Function	Indefinite Integral Formula
Power Rule ($n \neq -1$)	$\int x^n dx = \frac{x^{n+1}}{n+1} + C$
Reciprocal Rule	**$\int \frac{1}{x} \, dx = \ln\vert{}x\vert{} + C$**
Exponential (e^x)	$\int e^x dx = e^x + C$
Exponential (a^x)	$\int a^x dx = \frac{a^x}{\ln a} + C$
Sine Function	$\int \sin x dx = -\cos x + C$
Cosine Function	$\int \cos x dx = \sin x + C$

Secant Squared	$\int \sec^2 x dx = \tan x + C$
Standard Inverse Form	$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1}\left(\frac{x}{a}\right) + C$
Standard Rational Form	$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$

3. Core Integration Techniques

A. Integration by Substitution (U-Substitution)

- **Concept: Used when an integrand contains a composite function along with its derivative factor. By substituting $u = g(x)$, the integral simplifies into a standard form.**
- **Solved Numerical: Evaluate $\int 2x(x^2 + 5)^4 dx$.**
    - **Solution:**
        1. **Let $u = x^2 + 5$.**
        2. **Differentiate both sides: $du = 2x dx$.**
        3. **Substitute u and du into the integral:**

$$\int (x^2 + 5)^4 \cdot (2x dx) = \int u^4 du$$

4. **Integrate using the power rule:**

$$= \frac{u^5}{5} + C$$

5. **Substitute back $u = x^2 + 5$:**

$$= \frac{(x^2+5)^5}{5} + C$$

6. **(Answer)**

B. Integration by Parts

- **Concept: Derived from the product rule of differentiation. Used when the integrand is a product of two distinct types of functions (e.g., algebraic and logarithmic/trigonometric).**
- **Formula:**
    - $\int u \cdot v dx = u\int v dx - \int\left(\frac{du}{dx}\int v dx\right)dx$
- ***(Hint for choosing u: Use ILATE order: Inverse trig, Logarithmic, Algebraic, Trigonometric, Exponential).***
- **Solved Numerical: Evaluate $\int x \cdot e^x dx$.**
    - **Solution:**

1. **Choose $u = x$ (Algebraic) and $v = e^x$ (Exponential).**
2. **Compute derivatives/integrals: $\frac{du}{dx} = 1$ and $\int v dx = e^x$.**
3. **Apply the Integration by Parts formula:**

$$\int xe^x dx = x \cdot e^x - \int (1 \cdot e^x) dx$$

5.

$$= xe^x - e^x + C = e^x(x - 1) + C$$

7. **(Answer)**

C. Partial Fractions

- **Concept: Used to break down complex rational algebraic fractions into simpler, easily integrable fractions.**
- **Solved Numerical: Evaluate $\int \frac{1}{x^2 - 9} dx$.**
  - **Solution:**
    1. **Factorize the denominator: $x^2 - 9 = (x - 3)(x + 3)$.**
    2. **Express as partial fractions:**

$$\frac{1}{(x-3)(x+3)} = \frac{A}{x-3} + \frac{B}{x+3}$$

    4. **Solve for constants A and B:**

$$1 = A(x + 3) + B(x - 3)$$

       - **Put $x = 3 \implies 1 = 6A \implies A = \frac{1}{6}$**
       - **Put $x = -3 \implies 1 = -6B \implies B = -\frac{1}{6}$**
    6. **Integrate each term separately:**

4. Definite Integrals & Fundamental Theorem of Calculus

- **Fundamental Theorem of Calculus (Part 2): If f is continuous on $[a, b]$ and F is any antiderivative of f, then:**
  - $$\int_a^b f(x) dx = F(b) - F(a)$$
- **Solved Numerical: Evaluate $\int_0^2 (3x^2 + 2x + 1) dx$.**
  - **Solution:**
    1. **Find the indefinite integral:**

$$F(x) = \left[\frac{3x^3}{3} + \frac{2x^2}{2} + x\right] = [x^3 + x^2 + x]$$

    3. **Apply upper and lower limits ($x = 2$ and $x = 0$):**

$$F(2) = (2)^3 + (2)^2 + (2) = 8 + 4 + 2 = 14$$

5.

$$F(0) = (0)^3 + (0)^2 + (0) = 0$$

7. **Subtract:**

8. $\int_0^2 (3x^2 + 2x + 1)dx = 14 - 0 = 14$

VECTOR CALCULUS

PART 1: VECTORS IN PLANE & SPACE, LINEAR INDEPENDENCE

1. Vectors in a Plane and Space

- **Deep Definition:** A **vector** is a mathematical quantity possessing both magnitude and direction, represented geometrically by a directed line segment.
    - In a **2D Plane**, a vector is expressed as $\vec{A} = x\hat{i} + y\hat{j}$ or as an ordered pair (x, y).
    - In 3D **Space**, a vector includes three spatial components:
        - $\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$
    - **Magnitude (Length):** $\vert{}\vec{A}\vert{} = \sqrt{A_x^2 + A_y^2 + A_z^2}$
    - **Unit Vector:** A vector of magnitude 1 in the direction of $\vec{A}$, given by $\hat{u} = \frac{\vec{A}}{\vert{}\vec{A}\vert{}}$.

2. Linear Dependence and Independence of Vectors

- **Deep Definition:** A set of vectors $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ is said to be **linearly independent** if no vector in the set can be expressed as a linear combination of the others. Mathematically, the vector equation:
  $$c_1\vec{v}_1 + c_2\vec{v}_2 + \dots + c_n\vec{v}_n = 0$$
  has only the trivial solution where all scalars $c_1 = c_2 = \dots = c_n = 0$. If non-zero scalars exist that satisfy this equation, the vectors are **linearly dependent**.
- **Determinant Test for 3D Vectors:** Three 3D vectors $\vec{A}, \vec{B}, \vec{C}$ are coplanar and linearly dependent if and only if their scalar triple product (determinant formed by their components) equals zero:
    - $\det[\vec{A}\vec{B}\vec{C}] = 0$

PART 2: DOT PRODUCT AND CROSS PRODUCT OF VECTORS

1. Dot Product (Scalar Product)

- **Definition:** The dot product of two vectors $\vec{A}$ and $\vec{B}$ yields a scalar quantity:
  $$\vec{A} \cdot \vec{B} = \vert{}\vec{A}\vert{} \vert{}\vec{B}\vert{} \cos\theta$$
  where θ is the angle between the vectors.
- **Component Form:** If $\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$ and $\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$:
    - $\vec{A} \cdot \vec{B} = A_xB_x + A_yB_y + A_zB_z$
- **Orthogonality Condition:** Two non-zero vectors are perpendicular (orthogonal) if and only if $\vec{A} \cdot \vec{B} = 0$.

2. Cross Product (Vector Product)

- **Definition:** The cross product of two vectors $\vec{A}$ and $\vec{B}$ yields a new vector perpendicular to both, given by:
  $$\vec{A} \times \vec{B} = \vert{}\vec{A}\vert{} \vert{}\vec{B}\vert{} \sin\theta \, \hat{n}$$
  where $\hat{n}$ is a unit vector perpendicular to the plane containing $\vec{A}$ and $\vec{B}$ (following the right-hand rule).
- **Determinant Form:**
  $$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$
- **Parallelism Condition:** Two non-zero vectors are parallel if and only if $\vec{A} \times \vec{B} = 0$.
- **Solved Numerical:** Find $\vec{A} \times \vec{B}$ given $\vec{A} = 2\hat{i} - 3\hat{j} + \hat{k}$ and $\vec{B} = \hat{i} + 4\hat{j} - 2\hat{k}$.
  - **Solution:**
    $$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -3 & 1 \\ 1 & 4 & -2 \end{vmatrix}$$
  - $$= \hat{i}[(-3)(-2) - (1)(4)] - \hat{j}[(2)(-2) - (1)(1)] + \hat{k}[(2)(4) - (-3)(1)]$$
  - $$= \hat{i}(6 - 4) - \hat{j}(-4 - 1) + \hat{k}(8 + 3)$$
  - $$= 2\hat{i} + 5\hat{j} + 11\hat{k}$$
  - **(Answer)**

PART 3: VECTOR DIFFERENTIAL OPERATORS (GRADIENT, DIVERGENCE, CURL)

The vector differential operator, known as **Del (∇)**, is defined in Cartesian coordinates as:

$$\nabla = \hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}$$

1. Gradient of a Scalar Field

- **Definition:** The gradient operates on a scalar field $\phi(x, y, z)$ to produce a vector field.
  - $Grad(\phi) = \nabla\phi = \hat{i}\frac{\partial\phi}{\partial x} + \hat{j}\frac{\partial\phi}{\partial y} + \hat{k}\frac{\partial\phi}{\partial z}$
- **Solved Numerical:** Find the gradient of $\phi(x, y, z) = 3x^2y - y^3z^2$ at the point $(1, -2, 1)$.
  - **Solution:**
    1. Compute partial derivatives:
       - $\frac{\partial\phi}{\partial x} = \frac{\partial}{\partial x}(3x^2y - y^3z^2) = 6xy$
       - $\frac{\partial\phi}{\partial y} = \frac{\partial}{\partial y}(3x^2y - y^3z^2) = 3x^2 - 3y^2z^2$
       - $\frac{\partial\phi}{\partial z} = \frac{\partial}{\partial z}(3x^2y - y^3z^2) = -2y^3z$
    2. Construct $\nabla\phi$:
    3. $\nabla\phi = 6xy\hat{i} + (3x^2 - 3y^2z^2)\,\hat{j} - 2y^3z\hat{k}$
    4. Evaluate at point $(1, -2, 1)$:
    5. $\nabla\phi = 6(1)(-2)\,\hat{i} + (3(1)^2 - 3(-2)^2(1)^2)\,\hat{j} - 2(-2)^3(1)\,\hat{k}$
    6.
    7. $= -12\hat{i} + (3 - 12)\,\hat{j} - 2(-8)\,\hat{k} = -12\hat{i} - 9\hat{j} + 16\hat{k}$
    8. **(Answer)**

2. Divergence of a Vector Field

- **Definition:** The divergence of a vector field $\vec{V} = V_1\hat{i} + V_2\hat{j} + V_3\hat{k}$ is defined as the dot product of ∇ and $\vec{V}$, resulting in a **scalar** quantity:

$$Div(\vec{V}) = \nabla \cdot \vec{V} = \frac{\partial V_1}{\partial x} + \frac{\partial V_2}{\partial y} + \frac{\partial V_3}{\partial z}$$

- **Solenoidal Vector:** A vector field is called **solenoidal** if its divergence is identically zero ($\nabla \cdot \vec{V} = 0$).

3. Curl of a Vector Field

- **Definition:** The curl of a vector field $\vec{V}$ is defined as the cross product of ∇ and $\vec{V}$, resulting in a **vector** quantity:
  $$\text{Curl}(\vec{V}) = \nabla \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_1 & V_2 & V_3 \end{vmatrix}$$
- **Irrotational Vector:** A vector field is called **irrotational** if its curl is identically zero ($\nabla \times \vec{V} = 0$).

PART 4: PHYSICAL INTERPRETATION OF GRADIENT, DIVERGENCE, AND CURL

Operator	Mathematical Form	Physical Interpretation
Gradient	$\nabla\phi$ (Scalar → Vector)	Points in the direction of the **maximum rate of increase** of a scalar field, and its magnitude represents that maximum rate of change. It is always perpendicular (normal) to the level surfaces of ϕ.
Divergence	$\nabla \cdot \vec{V}$ (Vector → Scalar)	Measures the **net outward flux** or rate of expansion per unit volume from a given point. If positive, it acts as a **source** (fluid flowing outward); if negative, it acts as a **sink** (fluid flowing inward).
Curl	$\nabla \times \vec{V}$ (Vector → Vector)	Measures the **tendency of rotation** or angular circulation of a vector field about a point. If non-zero, it indicates a vortex or swirling motion around an axis aligned with the curl vector.